Web and Mobile Platform for Scientific Conference Submission Management

Multidisciplinary Project – 2nd Year

Academic Year 2026–2027

Project Details

Project Type:	2nd Year Multidisciplinary Project
Academic Year:	2026–2027
Number of Students:	2–4 students
Institution:	École Supérieure en Informatique – Sidi Bel Abbès

1. Project Context

Scientific conferences need efficient management of paper submissions, authors, reviewers, evaluation reports, decisions, and conference-related information. A dedicated digital platform can simplify these activities and provide a centralized environment for conference organizers, authors, and reviewers.

This project aims to develop a web and mobile platform for managing the scientific paper submission process of an academic conference. The system will support the complete submission workflow, from paper submission to review assignment and notification of decisions.

The project will provide students with practical experience in software engineering, web development, mobile development, databases, API development, user-interface design, and collaborative project management.

2. Project Objective

The main objective is to design and implement a web and mobile platform supporting the main activities involved in scientific conference paper submission and review management.

The platform should provide conference organizers with an organized environment for managing submissions and reviewers, while allowing authors to submit papers and monitor their submissions.

3. Specific Objectives

The specific objectives are:

1. Analyze the workflow of scientific conference paper submission.
2. Identify the roles involved in conference management.

3. Design a database for conferences, users, papers, reviews, and decisions.
4. Develop a web application for conference management.
5. Develop a mobile application providing selected functionalities.
6. Implement authentication and role-based access.
7. Provide paper submission and document-management functionalities.
8. Allow authors to manage their submissions.
9. Provide tools for conference organizers to manage submissions.
10. Provide reviewer assignment functionality.
11. Implement a basic paper-review workflow.
12. Provide decision-management functionality.
13. Provide notifications for important conference events.
14. Provide dashboards for conference organizers and authors.
15. Test and document the developed system.

4. Target Users

The platform will support:

- **Author:** submits papers and monitors their submissions.
- **Reviewer:** evaluates assigned papers.
- **Program Committee Member:** participates in the evaluation process.
- **Conference Organizer:** manages the conference and submissions.
- **Conference Administrator:** manages users, configuration, and system settings.

5. Main Functionalities

5.1. Conference Management

Organizers should be able to:

- Create a conference.
- Define conference information.
- Define important dates.
- Define submission topics.
- Manage conference status.
- Manage the list of reviewers and committee members.

5.2. Author Module

Authors should be able to:

- Create an account.
- Create a paper submission.
- Enter paper metadata.
- Add co-authors.
- Select appropriate conference topics.
- Upload the paper.
- Modify a submission before the deadline.
- Monitor the status of the submission.
- Receive notifications.

5.3. Reviewer Module

Reviewers should be able to:

- View review assignments.
- Accept or decline an assignment.
- Access assigned papers.
- Submit evaluation reports.
- Provide scores and recommendations.
- Monitor completed assignments.

5.4. Organizer Module

Conference organizers should be able to:

- View all submissions.
- Filter and search submissions.
- Manage authors.
- Manage reviewers.
- Assign reviewers to papers.
- Monitor review progress.
- Record paper decisions.
- Notify authors.
- Generate basic submission statistics.

6. Conference Submission Process

The platform should support a simplified process:

$$\text{Call for Papers} \rightarrow \text{Paper Submission} \rightarrow \text{Initial Check} \rightarrow \text{Reviewer Assignment} \rightarrow$$
$$\text{Review} \rightarrow \text{Decision} \rightarrow \text{Final Version}$$

Possible paper statuses include:

- Draft.
- Submitted.
- Under Initial Check.
- Under Review.
- Revision Needed.
- Accepted.
- Rejected.
- Camera-Ready.

7. Web and Mobile Applications

7.1. Web Application

The web application will provide the main environment for conference organizers, authors, reviewers, and program committee members.

It should provide complete access to the principal functions of the platform.

7.2. Mobile Application

The mobile application will provide selected functions such as:

- User authentication.
- Submission-status monitoring.
- Important conference notifications.
- Review assignment notifications.
- Review-status monitoring.
- Access to important conference information.

The mobile application may provide a simplified version of the functionalities available on the web platform.

8. System Architecture

The system should be designed using a modular architecture consisting of:

- Web interface.
- Mobile interface.
- Backend application.
- REST API.
- Database.
- Authentication and authorization services.
- File-management component.
- Notification component.

The web and mobile applications should communicate with a common backend through a well-defined API.

9. Database Design

The database may contain entities such as:

- Users, Roles, Conferences, Conference editions, Topics, Papers, Authors, Paper authors, Reviewers, Review assignments, Reviews, Decisions, Important dates, Notifications, Submission status history.

Students should design the database model before implementation.

10. Proposed Technologies

Possible technologies include:

- HTML, CSS, and JavaScript.
- React, Angular, or Vue for web development.
- Flutter or React Native for mobile development.
- Node.js, Django, Spring Boot, Laravel, or another suitable backend framework.
- MySQL or PostgreSQL.
- REST API.
- Git and GitHub/GitLab.

The project team may select alternative technologies according to its skills and project constraints.

11. Expected Deliverables

The team should provide:

1. Specifications document.
2. UML diagrams.
3. System architecture.
4. Database conceptual and logical models.
5. Functional web application.
6. Functional mobile application.
7. Backend application.
8. REST API.
9. Database implementation.
10. Authentication and role-management system.
11. Conference-management module.
12. Paper-submission module.
13. Reviewer-management module.
14. Basic review-management workflow.
15. Notification system.
16. Testing report.
17. Source code.
18. Technical documentation.
19. User guide.
20. Final project report.
21. Final demonstration.

12. Skills and Competencies

Students will develop practical skills in:

- System Specification and Design.
- UML modeling.
- Database design.
- Web development.
- Mobile development.
- Backend development.
- REST API development.
- Authentication and authorization.
- Software testing.
- Collaborative software development.
- Version control.
- Technical documentation.

13. Possible Future Extensions

The platform may later be extended with:

- Automatic paper-format validation, Reviewer recommendation, Conflict-of-interest management, Topic-based reviewer assignment, Automated email communication, Online payment integration, Registration management, Attendance management, Certificate generation, Conference program generation, Advanced conference statistics.

14. Keywords

Scientific Conference; Paper Submission; Peer Review; Conference Management; Web Application; Mobile Application; REST API; Database; Software Engineering.

Contact

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